UNIVERSITI TEKNOLOGI MARA (UiTM)

FACULTY OF ELECTRICAL ENGINEERING

EEE640 — SELECTED TOPICS FOR ELECTRONICS ENGINEERING

ASSIGNMENT 3 (50%) — INDIVIDUAL DESIGN PROJECT

GRID-CONNECTED PHOTOVOLTAIC SYSTEM DESIGN

Design of a 2.64 kWp Residential Rooftop GCPV System for a Three-Phase House in Taman Warisan Puteri, Seremban

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The selected residential roof at Taman Warisan Puteri, Seremban (2.7494° N, 101.9766° E).

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1. INTRODUCTION AND EXECUTIVE SUMMARY

This report presents the complete design of a residential grid-connected photovoltaic (GCPV) system for an existing house in Taman Warisan Puteri, Seremban, Negeri Sembilan. In line with the project brief, the governing design objective is to maximise the useful annual electrical energy harvested from the measured roof plane, while ensuring that the array remains electrically compatible with the existing three-phase TNB supply and with the selected inverter across the full range of operating temperatures expected at the site.

The design follows the standard GCPV sizing methodology taught in EEE640: site assessment, component selection, roof-area-constrained array sizing, inverter voltage-window and current matching, and energy-yield estimation using realistic derating factors. A three-phase string-inverter architecture was adopted because the premise is supplied by a three-phase service, as confirmed by the installed TNB smart meter (Section 2).

The roof measurements supplied by the homeowner give a usable plane of 4.17 m × 3.85 m (polygon area 16.54 m², perimeter 16.35 m). Because the roof area is limited, a compact high-efficiency N-type module is preferred so that the greatest possible capacity fits on the available surface. Six Trina Solar Vertex S+ TSM-440NEG9R.28 modules are arranged as a single 3 × 2 array, giving 2.64 kWp, connected to a Huawei SUN2000-3KTL-M1 three-phase inverter. The principal outcomes of the design are summarised below.

Table 1.1 : Summary of key design results.

Design Outcome	**Result**
Site	Residential house, Taman Warisan Puteri, Seremban, Negeri Sembilan (2.7494° N, 101.9766° E)
Usable roof plane	4.17 m × 3.85 m, measured area 16.54 m²
PV module selected	Trina Solar Vertex S+ TSM-440NEG9R.28, 440 Wp monocrystalline N-type i-TOPCon
Inverter selected	Huawei SUN2000-3KTL-M1, 3 kW, three-phase, 2 MPPT
Array configuration	1 string × 6 modules (Np = 1, Ns = 6),6 modules total
Installed DC capacity	2.64 kWp (DC/AC ratio = 0.88)
Estimated annual energy	≈ 3,333 kWh per year
Specific yield	≈ 1,263 kWh/kWp per year
Performance ratio (PR)	≈ 0.82 (82.4 %)
Grid interface	Three-phase Net Energy Metering (NEM) via TNB bidirectional meter
CO_2 avoided	≈ 1.95 tonnes CO_2 per year (≈ 49 tonnes over 25 years)

2. SITE IDENTIFICATION AND SITE INFORMATION

Accurate site information is the foundation of any GCPV design because it fixes the solar resource, the operating-temperature limits, the usable mounting area and the grid-connection arrangement. The selected premise is an occupied residential house in Seremban, Negeri Sembilan, its details are consolidated below.

2.1 Location and Premise

Table 2.1: Premise and electrical-supply information.

Parameter	Description
Premise type	Residential house
Location	Taman Warisan Puteri, Seremban, Negeri Sembilan, Malaysia
Geographic coordinates	2.7494° N latitude, 101.9766° E longitude
Roof measurement	Width 4.17 m,length 3.85 m,polygon area 16.54 m²,perimeter 16.35 m
Electrical supply	Three-phase, 3 × 230/400 V, 50 Hz (4-wire)
Metering scheme	Net Energy Metering (NEM) - existing TNB three-phase smart meter (Smart Meters Technologies APFC3-WC, 3×10(100) A, bidirectional capable)

2.2 Roof Assessment and Selected Plane

The measured roof plane is compact and approximately rectangular, making it suitable for a single, uniformly oriented array. Keeping all six modules on one plane as shown in Figure 2.2 at the same tilt and orientation avoids the current mismatch that would arise from splitting the array across differently facing roof faces, which keeps every module in the string matched and maximises the string output. The proposed physical arrangement of the 3 × 2 configuration on the measured plane is shown in Figure 2.1.

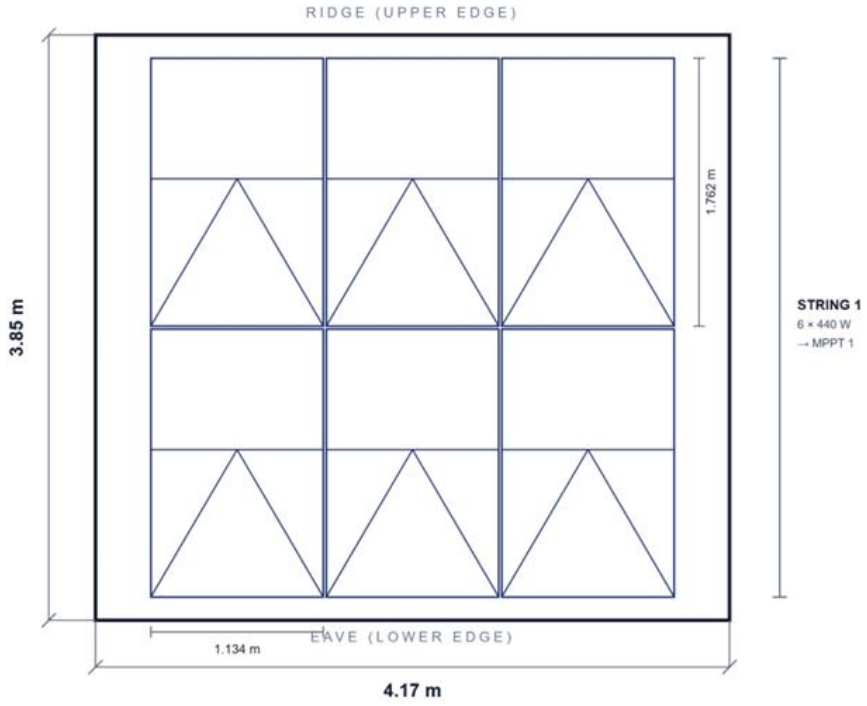


Figure 2.1: Proposed 3 × 2 PV module layout on the measured 4.17 m × 3.85 m roof plane.



Figure 2.2: Roof layout of the 6-module array(2 rows × 3 modules) on the selected roof

The single uniform array is adopted provided that a final on-site survey confirms structural adequacy, access clearance, ridge and edge setbacks, waterproof mounting points and the absence of significant near-field shading during the main generating window.

2.3 Existing TNB Meter (Three-Phase Confirmation)

The existing TNB smart meter at the premise is shown below in Figure 2.3. Its name plate confirms a three-phase, 4-wire service and it is a modern bidirectional capable smart meter suitable for the NEM scheme.

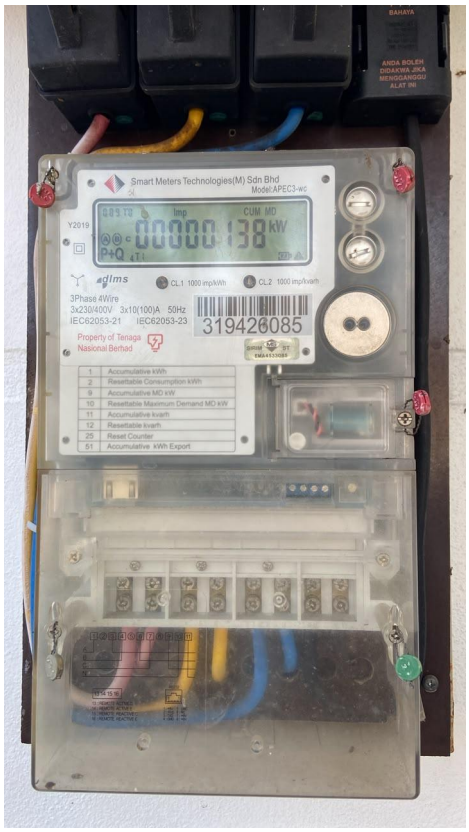


Figure 2.3: Existing TNB three-phase smart meter confirming the three-phase, four-wire service.

2.4 Solar Resource and Operating Conditions

Peninsular Malaysia is located near the equator and receives a good amount of solar radiation throughout the year. For the proposed GCPV system in Seremban, the average daily solar irradiation is taken as 4.2 kWh/m²/day, which is equivalent to 4.2 Peak Sun Hours (PSH). This value is considered suitable and is consistent with data from the Malaysian Meteorological Department (MetMalaysia), SEDA Malaysia, NASA POWER, and the Global Solar Atlas for the Negeri Sembilan region.

Since the site is located at a latitude of approximately 2.7° N, only a small tilt angle is required to achieve good solar energy collection. Therefore, the PV modules are installed flush with the existing tiled roof, which provides a near-optimum tilt angle while minimising self-shading between the modules.Module operating temperature is the single most important loss mechanism in a hot tropical climate, so conservative cell-temperature limits are adopted for the voltage-window checks, and an elevated average cell temperature is used for the energy estimate.

Table 2.2: Solar-resource and temperature design values.

Site / Design Parameter	Symbol	Value
Average daily irradiation (plane of array)	H_tilt	4.2 PSH/day (≈ 1,533 PSH/yr)
Average ambient temperature (generating hours)	T_amb,avg	30 °C
Average cell temperature (= T_amb,avg + 25)	T_cell,avg	55 °C
Minimum design cell temperature	T_cell,min	20 °C
Maximum design cell temperature	T_cell,max	75 °C
Standard Test Conditions	STC	1000 W/m², 25 °C, AM1.5

2.5 Design Assumptions

As permitted by the project brief, the following reasonable engineering assumptions are made and applied consistently throughout the report. The final installer must remeasure the sloped roof surface, confirm setbacks and walkway requirements, conduct a structural assessment and perform a detailed shading analysis before installation.

Table 2.3: Design assumptions and derating values used in the calculations.

Assumption	Value adopted
Usable roof area for the array	4.17 m × 3.85 m (rectangular, single plane)
Module-to-module installation gap	20 mm (0.02 m)
Plane-of-array irradiation	4.2 PSH/day
Module power tolerance (manufacturing mismatch)	0 / +5 W (k_mm = 1.00)
Soiling / dirt loss	5 % (k_dirt = 0.95)
First-year ageing / degradation	1 % (k_age = 0.99)
DC cable (array-to-inverter) efficiency	98 % (η_pv-inv = 0.98)
Significant shading	None assumed (k_shade not applied)

3. SELECTION OF PV MODULE AND INVERTER

3.1 PV Module Selection

Because the objective is to maximise energy from a fixed, limited roof area, a high-efficiency monocrystalline module is preferred: more watts per square metre means more installed capacity (and more energy) from the same roof. The Trina Solar Vertex S+ TSM-440NEG9R.28, a 440 Wp monocrystalline N-type i-TOPCon dual-glass module, was selected. Its compact 1762 mm × 1134 mm footprint allows all six modules to fit on the measured plane, while its high efficiency (≈ 22.0 %) and favourable temperature coefficient keep output high in the warm Seremban climate, where cells run well above 25 °C for most of the generating day. Its short-circuit current is moderate enough that a single string remains within the input-current limit of the selected residential inverter. Trina Solar is a Tier-1 manufacturer offering a long linear performance warranty, with low first-year degradation, and the module carries the IEC/MS certifications required for connection in Malaysia.

Table 3.1: Trina Solar Vertex S+ TSM-440NEG9R.28 representative datasheet values (STC).

Module parameter (at STC)	Symbol	Value
Maximum power	Pmp	440 W
Open-circuit voltage	Voc	52.2 V
Voltage at maximum power	Vmp	44.0 V
Short-circuit current	Isc	10.67 A
Current at maximum power	Imp	10.01 A
Temp. coefficient of P_mp	γPmp	- 0.30 %/°C
Temp. coefficient of V_oc	γVoc	- 0.24 %/°C
Module efficiency	η_mod	22.0 %
Dimensions (L × W × H)	-	1762 × 1134 × 30 mm
Cell technology	-	Dual-glass monocrystalline N-type i-TOPCon
Power tolerance	-	0 / +5 W
Maximum system voltage	-	1500 V DC

3.2 Inverter Selection

Because the existing supply is three-phase and the expected array size is in the 2.5–3 kW class, the Huawei SUN2000-3KTL-M1 is the smallest model in the three-phase family and that is the natural match. It is a transformer-less three-phase string inverter with two independent MPP trackers, a wide 140–980 V MPP voltage window, a 1100 V maximum DC input, and a high peak efficiency of 98.2 %. Its maximum input current of 13.5 A per MPPT comfortably accommodates one string of the selected module (I_{mp} = 10.01 A, I_{sc} = 10.67 A). One six-module string is connected to one MPPT, leaving the second MPPT available for a future expansion on a suitable additional roof plane. Huawei is an established brand with strong local service support, and the inverter complies with MS IEC 62109 safety and the grid-interconnection requirements enforced by the Energy Commission (ST) and TNB.

Table 3.2: Huawei SUN2000-3KTL-M1 representative datasheet values.

Inverter parameter	**Symbol**	**Value**
Rated AC output power	Pinv	3000 W
Nominal AC voltage / frequency	-	220/380 V – 230/400 V, three-phase (3W/N+PE), 50 Hz
Recommended maximum PV power	-	4500 Wp
Maximum DC input voltage	Vmax_inv	1100 V
Operating (MPP) voltage range	-	140 V – 980 V
Start-up voltage	-	200 V
Number of MPP trackers	-	2
Strings per MPP tracker	-	1
Maximum input current per MPPT	I_dc,max/MPPT	13.5 A
Maximum short-circuit current per MPPT	-	19.5 A
Maximum efficiency	η_inv	98.2 %
Protection / cooling	-	Anti-islanding, AFCI, RCMU, SPD,IP65, natural convection

3.3 Why a Three-Phase String Inverter

Because the premise is supplied by a three-phase service, a three-phase inverter is the correct match, it balances the exported PV power evenly across the three phases (L1, L2, L3), avoids the voltage-rise and single-phase loading problems that a single-phase inverter would create on a three-phase supply, and satisfies TNB's connection requirements for three-phase premises. A two-MPPT string inverter (rather than micro-inverters) is chosen so that the full DC string-sizing methodology which is series count, parallel count, voltage-window and current matching, it can be applied, while retaining a spare MPPT for future expansion. Maximum Power Point Tracking is referred to as MPPT. It is a sophisticated technological method used in solar charge controllers and inverters to continuously modify the solar panels' operational voltage and current. Below is an example of a Current-Voltage (I-V) Graph which shows the change in current when the panel voltage changes.

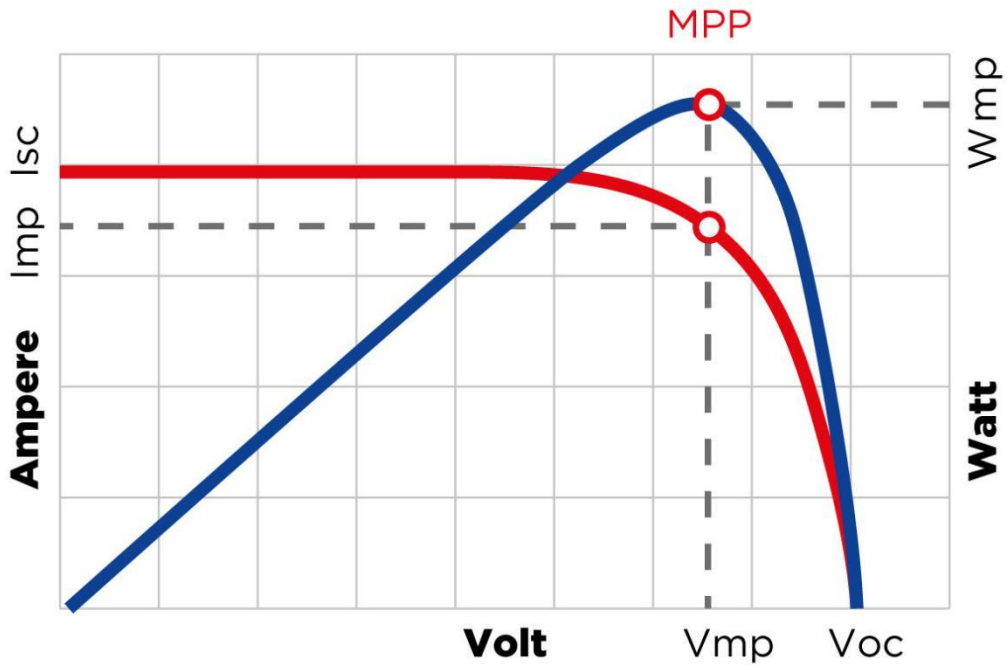


Figure 3.1: Current-Voltage (I-V) Graph

4. PV ARRAY CONFIGURATION AND SYSTEM SIZING

The array is sized using the standard GCPV design procedure. The roof area sets the maximum module count,the inverter rating sets the optimal count,and the inverter voltage window and input-current limit set the allowable number of modules in series (N_s) and strings in parallel (N_p). The notation r.d. denotes “round down” and r.u. denotes “round up” to the nearest whole module.

Step 1 : Selected components

PV module that selected :

Trina Solar Vertex S+ TSM-440NEG9R.28 as shown in **Table 3.1**.

Inverter that selected :

Huawei SUN2000-3KTL-M1 as shown in **Table 3.2.**

Step 2 : Maximum number of modules on the roof

With a usable roof rectangle of Wroof = 4.17 m and Lroof = 3.85 m, a module length Lmod = 1.762 m, width Wmod = 1.134 m, and an installation gap, lg = 0.02 m, both orientations are evaluated.

$$N_roof\ _max_LA = r.d.\left[\frac{Wroof}{Wmod\ +\ lg}\right] \times r.d.\left[\frac{Lroof}{Lmod\ +lg}\right]$$

$$N_roof\ _max_LA = r.d.\left[\frac{4.17}{1.134 +\ 0.02}\right] \times r.d.\left[\frac{3.85}{1.762 + 0.02}\right]$$

$$= r.d.\,[3.61] \times r.d.\,[2.16] =\ 3 \times 2 = 6\ modules$$

$$N_roof\ _max_LU = r.d.\left[\frac{Wroof}{Lmod\ +\ lg}\right] \times r.d.\left[\frac{Lroof}{Wmod\ +lg}\right]$$

$$N_roof\ _max_LA = r.d.\left[\frac{4.17}{1.762\ +\ 0.02}\right] \times r.d.\left[\frac{3.85}{1.134 + 0.02}\right]$$

$$= r.d.\,[2.34] \times r.d.\,[3.34] =\ 2 \times 3 = 6\ modules$$

Maximum modules that can be loaded by the roof: N_roof, maximum = 6 (both orientations). This arrangement uses all 6 modules for full and efficient use of the roof plane. The 3 × 2 arrangement is used for a simple rectangular layout and easier cable routing.

Step 3 : Optimal array size for the inverter

The optimal module count keeps the array power within the inverter's preferred sizing band, using sizing factors SF1 = 0.90 and SF2 = 1.00.

$$N_tot_opt = r.u.\left[\frac{Pinv}{SF2\ +\ Pmp_stc}\right] to\ r.d.\left[\frac{Pinv}{SF1 + Pmp_stc}\right]$$

$$N_tot_opt = r.u.\left[\frac{3000}{1.00\ +\ 440}\right] to\ r.d.\left[\frac{3000}{0.90 + 440}\right] =\ 7\ to\ 7\ modules$$

$$= r.u.\,[6.82] to\ r.d.\,[7.58] =\ 7\ to\ 7\ modules$$

The inverter that preferred count is 7 modules, however, the roof can physically accommodate only 6. The roof area is therefore the binding constraint. Six modules (2.64 kWp) are accepted because they lie within the inverter manufacturer's recommended PV power of 4500 Wp and give a practical DC/AC ratio of 0.88.

Step 4 : Revised inverter voltage limits

Safety margins are applied: λupper = 5 % on the maximum input voltage and λlower = 10 % on the minimum MPP voltage.

$$Vmax_inv_rev = (100\% - \lambda\text{upper})\ \times Vmax_inv$$

$$=\ (100\%\ -5\%) \times 1100 = 1045\ V$$

$$Vmin_win_inv_rev = (100\% +\ \lambda\text{lower})\ \times Vmin_win_inv$$

$$=\ (100\%\ +5\%) \times 140 = 145\ V$$

Step 5 : Module voltage extremes at the design temperatures

Maximum module Voc occurs at the minimum cell temperature while minimum Vmp at the maximum cell temperature, then reduced by a 5 % voltage-drop allowance.

$$Vmax_oc = Voc_stc\ \times (1 +\ \left[(\frac{\gamma\text{Voc}}{100}) \times (Tcell_max\ -\ Tstc)\right]$$

$$= 52.2\ \times (1 +\ \left[(\frac{-0.24}{100}) \times (20\ -\ 25)\right] = 52.83\ V$$

$$Vmin_mp = Vmp_stc\ \times (1 +\ \left[(\frac{\gamma\text{Vmp}}{100}) \times (Tcell_min\ -\ Tstc)\right]$$

$$= 44.0\ \times (1 +\ \left[(\frac{-0.30}{100}) \times (75\ -\ 25)\right] = 37.40\ V$$

$$Vmin_mp_vd = 0.95 \times\ Vmin_mp\ = 0.95\ \times 37.40\ = 35.53\ V$$

Step 6 : Allowable number of modules in series (N_s)

$$N_s, max_oc = r.d.\left(\frac{Vmax_inv_rev}{Vmax_oc}\right) = r.d.\left(\frac{1045}{52.83}\right) = 19$$

$$N_s, min_mp = r.u.\left(\frac{Vmin_win_inv_rev}{Vmin_mp_vd}\right) = r.u.\left(\frac{154}{35.53}\right) = 5$$

Series range: N_s = 5 to 19 modules per string. The selected 6-module string satisfies this comfortably. (The wide upper limit reflects the inverter's 1100 V DC input which is the roof area, not voltage that is the binding constraint.)

Step 7 : Allowable number of parallel strings (N_p)

The maximum number of parallel strings per MPP tracker is found by dividing the inverter's maximum input current per MPPT by the module short-circuit current, uprated by a safety factor ω = 25 %. The inverter has two MPP trackers, each rated 13.5 A, I_dc_max_per_mppt_inv.

For MPPT 1:

I_dc_max_per_mppt_inv = 13.5 A

$$N_p, max_per_mppt_inv = r.d.\left(\frac{I_dc_max_per_mppt_inv}{(100\% + \omega) \times \text{Isc_stc}}\right)$$

$$= r.d.\left(\frac{13.5}{(100\% + 25\%) \times 10.67}\right) = r.d.\left(\frac{13.5}{13.34}\right) = r.d.(1.012) = 1\ string$$

For MPPT 2:

I_dc_max_per_mppt_inv = 13.5 A

$$N_p, max_per_mppt_inv = r.d.\left(\frac{I_dc_max_per_mppt_inv}{(100\% + \omega) \times \text{I_sc_stc}}\right)$$

$$= r.d.\left(\frac{13.5}{(100\% + 25\%) \times 10.67}\right) = r.d.\left(\frac{13.5}{13.34}\right) = r.d.(1.012) = 1\ string$$

Therefore,

$$\text{N_p, max} = \sum_{i=1}^{k} (N_p, max_per_mppt_inv\ for\ MPPT\ i) = 1 + 1 + 2\ strings$$

The single-string current is also confirmed within the inverter limits:

- String operating current I_mp = 10.01 A < 13.5 A (max input current per MPPT)
- String short-circuit current I_sc = 10.67 A < 19.5 A (max short-circuit current per MPPT)

Step 8 : Candidate array configurations

Combining N_s ∈ [5, 19] with N_p ∈ [1, 2], and discarding any configuration that exceeds the roof limit of 6 modules, the realistic candidates are:

Table 4.1 : Candidate array configurations.

Number of parallel strings, N_p	Number of PV modules per string, N_s	Total number of PV modules, N_tot	Remark
1	6	6	Fits roof, uses full roof plane and one MPPT (SELECTED)
1	5	5	Fits roof but wastes one module space
2	3	6	Fits roof and splits over 2 MPPT but each string very short

Step 9 : Selected configuration

Each candidate is checked against three criteria: Criterion 1, it must fit within the roof capacity of 6 modules. Next Criterion 2, it should fall within the inverter's optimal sizing range and Criterion 3, where several options fit, the one giving the largest module count (hence the most energy) is preferred. Here the optimal range which is in Step 3 is 7 modules but the roof holds only 6, so no candidate can reach the optimal band like Criterion 1 (roof fit) therefore governs, and the largest roof-fitting option is chosen to maximise energy. The 1 × 6 arrangement fills the roof with all 6 modules as a single, cleanly routed string on one MPP tracker, leaving the second MPPT free for a future additional array.

Final array configuration: N_p = 1, N_s = 6, N_tot = 6 modules, it is arranged as three columns of two (3 × 2 landscape), connected as one 6-module string to MPPT 1 of the inverter.

4.1 Voltage and Current Verification of the Selected String

Table 4.2 : String voltage/current verification (6-module string)

Check (6-module string)	Computed	Inverter limit
String Voc at 20 °C (cold)	6 × 52.83 = 317.0 V	< 1100 V (and < 1045 V)
String Vmp at STC	6 × 44.0 = 264.0 V	within 140–980 V
String Vmp at 75 °C (after V-drop)	6 × 35.53 = 213.2 V	> 154 V
String operating current (I_mp)	10.01 A	< 13.5 A per MPPT
String short-circuit current (I_sc)	10.67 A	< 19.5 A per MPPT

5. ANNUAL ENERGY YIELD ESTIMATION

Step 10 : Installed array DC capacity

$$P_array, STC = N_tot \times Pmp = 6 \times 440 = 2640\, W \approx 2.64kW$$

Step 11 : Expected annual energy output

The expected energy is the array capacity multiplied by the plane of array sun hours and by every loss (derating) factor between the cells and the grid connection. The temperature derate uses the average cell temperature of 55 °C from Table 2.2.

$$k_temp, Pmp = 1 + \left[\left(\frac{\gamma \mathrm{Vmp}}{100}\right) \times (T_cell, avg - Tstc)\right]$$

$$= 1 + \left[\left(\frac{-0.30}{100}\right) \times (55 - 25)\right] = 0.91$$

Table 5.1: Derating factors applied in the energy estimate.

Derating factor	Symbol	Value
Module mismatch / power tolerance	k_mm	1.00
Temperature (cell at 55 °C)	k_temp,Pmp	0.91
Soiling / dirt	k_dirt	0.95
Ageing (first year)	k_age	0.99
DC cabling (array → inverter)	η_pv-inv	0.98
Inverter conversion	η_inv	0.982

Thus, annual expected energy:

$$\mathrm{E_sys, annual} = \mathrm{P_array} \times \mathrm{H_tilt} \times \mathrm{k_temp, mm} \times \mathrm{k_temp} \times \mathrm{k_dirt} \times \mathrm{k_age} \times \mathrm{\eta_pv_inv} \times \mathrm{\eta_inv}$$

$$= 2.64 \times 1533 \times 1.00 \times 0.91 \times 0.95 \times 0.99 \times 0.98 \times 0.982 \approx 3{,}333\ \mathrm{kWh/year}$$

Step 12 : Specific yield

$$SY = \frac{\frac{Esys, annual\ exp}{1000}}{\frac{P_array, STC}{1000}}$$

$$SY = \frac{\frac{3{,}333kWh}{1000}}{\frac{2.64kW}{1000}} = 1{,}263\frac{kWh}{kWp}$$

Step 13 : Performance ratio

$$PR = k_mm \times k_temp, Pmp \times k_dirt \times k_age \times \eta_pv_inv \times \eta_inv$$
$$= 1.00 \times 0.91 \times 0.95 \times 0.99 \times 0.98 \times 0.982 \approx 0.824 \approx 82.4\ \%$$

A performance ratio of about 0.82 is a healthy value for a rooftop system in a hot tropical climate, where the temperature loss alone accounts for roughly 9 % of the deficit from unity.

5.1 Environmental Energy Summary

Using an indicative Malaysian grid emission factor of about 0.585 kg CO_2 per kWh, the avoided emissions are:

$$CO_2 \text{ avoided} = E_sys, annual \times 0.585 = 3333 \times 0.585 \approx 1{,}950 \text{ kg/yr} \approx 1.95 \text{ tonnes/yr}$$

The proposed GCPV system is expected to reduce approximately 1.95 tonnes of CO_2 emissions annually. Over its estimated service life of 25 years, this equates to approximately 49 tonnes of CO_2 emissions avoided, making the system beneficial to the environment in the long term. Under the three-phase Net Energy Measurement (NEM) scheme, the electricity generated by the PV system is first used to meet household electricity demand, helping to reduce monthly electricity bills. Based on an average domestic electricity tariff of RM0.50 per kWh, the system can provide an estimated annual saving of around RM1,650. This means that the installation cost can typically be recovered within 6 to 8 years. After that, the electricity generated by the PV system can be considered virtually free for the entire service life of the PV modules. The CO_2 emission reductions presented in this report are based on the current Malaysian grid emission factors. Therefore, the values should be updated if newer official grid emission factors are published for the reporting year.

Table 5.2: Key performance summary.

Performance index	Symbol	Result
Installed array capacity	P_array,STC	2.64 kWp
Estimated annual energy	E_sys,annual	≈ 3,333 kWh/yr
Average daily energy	-	≈ 9.1 kWh/day
Specific yield	SY	≈ 1,263 kWh/kWp
Performance ratio	PR	≈ 82.4 %
CO_2 avoided	-	≈ 1.95 t/yr

6. ELECTRICAL SCHEMATIC OF THE DESIGN

The single-line block diagram below in Figure 6.1 shows the complete electrical design from the PV array to the grid connection, including all protection and isolation devices on both the DC and AC sides also shown in Figure 6.2 which is the electrical schematic of the system design . Power flows from the 6-module string, through DC isolation, fusing and surge protection, into the MPP input of the Huawei inverter,the three-phase inverter output then passes through AC isolation, surge protection and an AC circuit-breaker to the existing three-phase main distribution board, and the TNB bidirectional (NEM) meter, from where it supplies the household loads and exports any surplus to the grid.

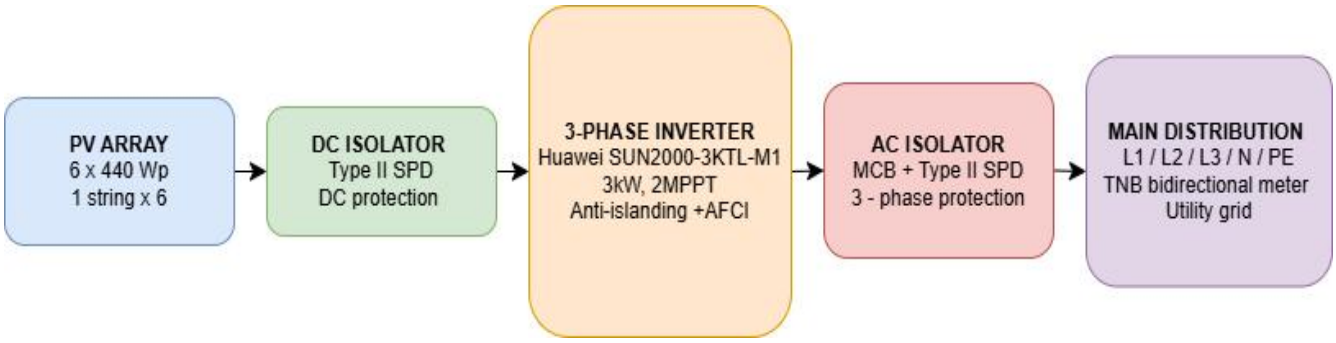


Figure 6.1: Single line electrical block diagram of three-phase grid connected PV system.

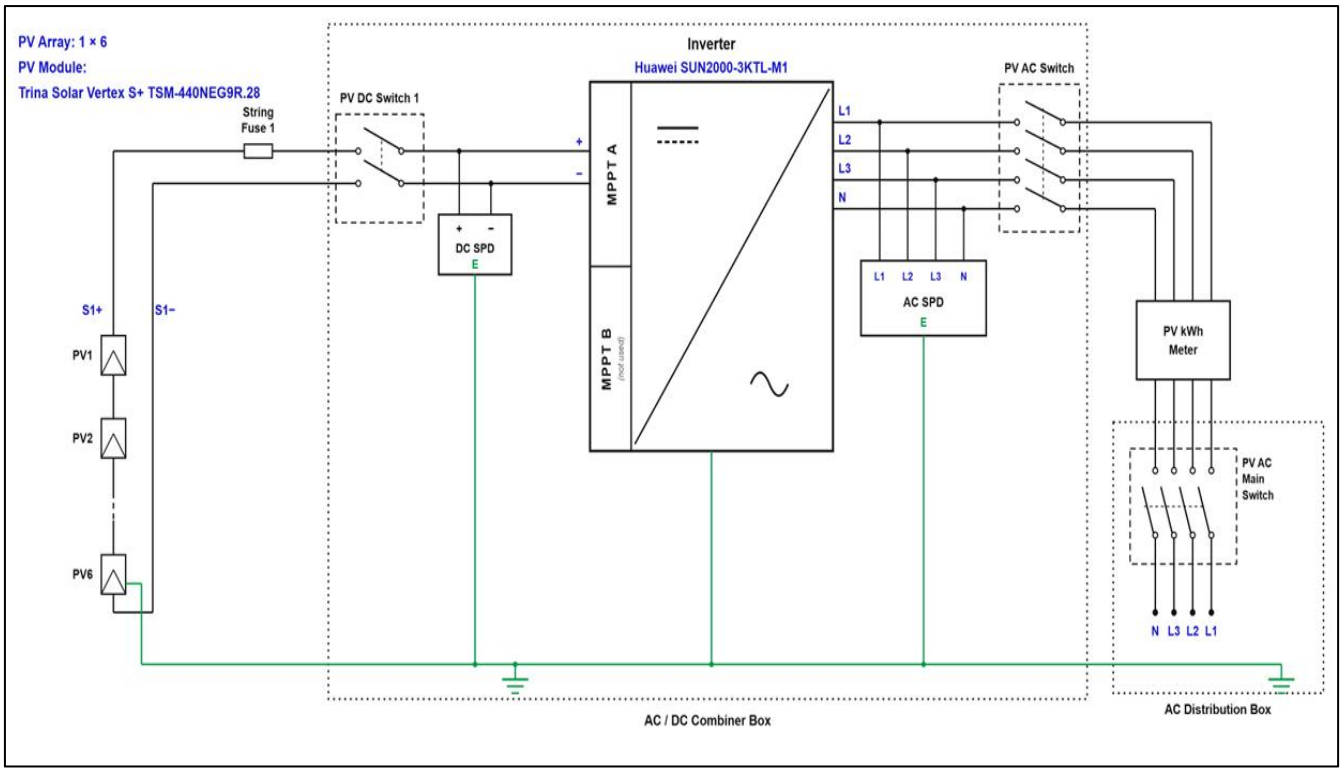


Figure 6.2: Single line electrical schematic of the 2.64 kWp three-phase grid-connected PV system.

6.1 Balance of System Components

The balance of system (BOS) components are selected in accordance with MS IEC 60364 / MS 1837 (PV installation requirements) and the protection principles taught in the course.

Table 6.1: Balance-of-system components and their functions.

Component	Specification	Function
PV modules	Trina Vertex S+ TSM-440NEG9R.28, 440 Wp × 6	DC power generation
Inverter	Huawei SUN2000-3KTL-M1, 3 kW, 3-phase, 2 MPPT	DC→AC conversion, MPPT, grid synchronisation
Mounting	Anodised-aluminium rails + SS304 clamps, tile-roof hooks	Secures array to the tiled roof
DC cable	4 mm² Cu, PV-rated, 1500 V DC, double-insulated	String to inverter
AC cable	4-core 4 mm² Cu / PVC (3φ + N)	Inverter to main DB
DC isolator	1000 V DC, ≥ 20 A, lockable	Safe DC disconnection
DC SPD	Type 2, U_c ≥ 600 V DC	Surge / lightning protection (DC side)
AC SPD	Type 2, 40 kA, 3-phase 400 V	Surge protection (AC side)
AC MCB	3-pole, 16 A, Type C	AC over-current protection & isolation
Earthing	10 mm² Cu PE conductor + earth electrode	Equipotential bonding & safety earthing
Labels / signage	PV DC warning, dual-supply & isolation labels	Safety identification for operators
Metering	TNB three-phase bidirectional smart meter	NEM import/export measurement

Cable cross sections are chosen so that the voltage drop stays within the limits of MS IEC 60364 (typically below 3 % on the DC side and below 2 % on the AC side) for the expected run lengths of roughly 15–25 m. On a three-phase 400 V supply the inverter's rated AC current is only about $3000\ \mathrm{W} / (\sqrt{3} \times 400\ \mathrm{V}) \approx 4.3\ \mathrm{A}$ per phase, so a 16 A three-pole MCB provides ample margin, and the DC SPD continuous voltage (U_c) exceeds the cold condition string Voc of about 317 V.

7. SUSTAINABILITY, SOCIETAL, HEALTH, SAFETY, LEGAL AND CULTURAL IMPACT

This section evaluates the proposed GCPV system as a sustainable engineering solution. It first proposes the specific sustainable materials, systems and processes adopted in the design, then identifies the sustainability goals the design accommodates, and finally explains the wider impact of the system on society, the environment and sustainable development, together with the relevant health, safety, legal and cultural considerations.

7.1 Proposed Sustainable Materials, Systems and Processes

Three distinct sustainable elements are proposed and justified, each considered in both its environmental and its societal context.

(i) High-efficiency monocrystalline N-type i-TOPCon PV modules

Environmental justification: The Trina Vertex S+ TSM-440NEG9R.28 is a conversion of sunlight, which is an inexhaustible and indigenous source. Eliminates the need for fuel, combustion, operational emissions while converting directly to electricity without a grid generation system. High efficiency (≈ 22.0 %) and modern N-type i-TOPCon cell technology translates to more energy produced per square metre and per unit of manufacturing material, and low temperature coefficient to maintain high production volume in Seremban, minimising the embodied energy per kWh generated and therefore the roof space footprint. Typically, the energy expended in producing the module is recovered again in one to two years, and the module has a service life of 25 years or more. The dual glass module can be recycled to a great extent, the aluminium frame, glass and silicon can be recovered at end of life and contribute to a circular economy and reduce the landfill.

Societal justification: A long linear performance warranty, that is a good Tier-1 module, provides the household with a long-term, low-maintenance clean energy supply, which will enhance the energy security of the family and shield them from rising electricity tariffs. If efficient modules can be adopted more widely, then the nation can work towards its renewable energy goals and continue to have rooftops as the energy source, which is more socially preferred, than greenfield land.

(ii) Grid-connected NEM architecture with a high-efficiency three-phase smart inverter

Environmental consideration: No batteries are required in the grid connected NEM design, eliminating battery manufacturing and disposal issues. The excess energy produced during the day is sold to the grid and then credited back, meaning that no energy is wasted in curtailments and every

bit produced is put to good use in maximizing the environmental benefit of every panel. The 98.2 % efficient transformer less inverter reduces the conversion losses while the dual MPPT tracking optimises the available energy under all conditions. It is a 3 phase inverter so the export is evenly distributed in each phase, thus minimizing loss and voltage imbalance.

Societal justification: NEM helps to change the role of consumers into good consumers by reduces the cost of electricity, increases speed of payback and makes clean energy financially available to ordinary families. The system can inject balanced excess power back into the local three-phase grid at peak demand times of day, which promotes grid stability and helps delay expensive power generation and grid expansion costs that benefit the entire community. The integrated monitoring aspect of the inverter also increases the awareness of the household of the use of energy.

(iii) Recyclable aluminium mounting and low-loss copper BOS with full surge and earthing protection

Environmental justification: Anodised aluminium is chosen for the mounting structure because it does not rust easily and can withstand Malaysia's hot and humid weather. It also lasts for many years and can be recycled after its service life, making it a sustainable material. Since the solar panels are installed on the existing roof, no additional land or major construction is needed. Properly sized copper cables are selected to reduce energy loss and improve the overall efficiency of the PV system.

Societal justification: Seremban often experiences thunderstorms, so installing Type 2 surge protection on both the DC and AC sides is important to protect the inverter, household appliances and the PV system from lightning and electrical surges. Together with a proper earthing system, it improves electrical safety for the homeowner and makes the system more reliable. This helps increase public confidence in adopting rooftop solar energy.

7.2 Sustainability Goals (SDGs) Accommodated

Table 7.1: Sustainability goals accommodated by the design.

Sustainability Goal	Justification
SDG 7 — Affordable & Clean Energy	Generates ≈ 3,333 kWh/yr of clean renewable electricity on-site and lowers the household's energy cost through NEM, increasing the share of affordable clean energy.
SDG 13 — Climate Action	Avoids ≈ 1.95 tonnes of CO2 per year by displacing fossil fuelled grid power.
SDG 11 — Sustainable Cities & Communities	Converts unused roof space into a clean-generation asset, making the home and community more sustainable and resilient without consuming additional land.
SDG 12 — Responsible Consumption & Production	Battery-free, self-consumption-plus-export design with recyclable modules and aluminium mounting promotes responsible energy use and a circular material flow.
SDG 9 — Industry, Innovation & Infrastructure	Adopts modern N-type i-TOPCon and smart-inverter technology and adds distributed clean-energy infrastructure to the existing grid, supporting Malaysia's green-technology industry.

7.3 Impact on Society, Environment and Sustainable Development

Societal impact: The proposed GCPV system helps reduce the household's monthly electricity bill by generating clean electricity from solar energy. It also reduces the impact of future electricity tariff increases, allowing homeowners to save more money for other daily expenses. As more houses install rooftop solar systems, it encourages the local community to adopt renewable energy. In addition, the installation and maintenance of solar PV systems create job opportunities for local engineers, technicians and installers, contributing to the growth of the green energy sector.

Environmental impact: The GCPV system produces electricity without emitting greenhouse gases, noise or air pollution during operation. Based on the estimated annual energy generation, the system can reduce approximately 1.95 tonnes of CO2 emissions per year by replacing electricity generated from fossil fuels. This also helps reduce other harmful pollutants such as sulphur oxides (SOx), nitrogen oxides (NOx) and particulate matter, which can affect air quality and human health. Furthermore, the selected PV modules and aluminium mounting structure are recyclable, helping to reduce environmental impact throughout the system's lifetime.

Sustainable-development impact: It has demonstrated how renewable energy can be integrated into residential buildings to achieve a balance between economic, environmental and social benefits. The system allows homeowners to reduce electricity costs while using clean, environmentally friendly energy. It also supports Malaysia's renewable energy and carbon reduction targets, while encouraging the wider adoption of rooftop solar systems.

7.4 Health and Safety

Safety is one of the most important considerations in this GCPV system design. DC and AC isolators are installed to allow the system to be safely disconnected during maintenance. Type 2 surge protection devices (SPD) are installed on both the DC and AC sides to protect the PV system from lightning strikes and electrical surges. A correctly rated three-pole AC miniature circuit breaker (MCB) is also used to provide overcurrent protection. In addition, a 10 mm^2 copper earthing conductor and earth electrode are installed to ensure all metal parts are properly grounded, reducing the risk of electric shock and allowing faults to be cleared quickly.

The selected inverter also includes anti-islanding protection, which automatically disconnects the PV system when there is a power outage. This protects utility workers who may be carrying out maintenance on the electrical network. The inverter is also equipped with Arc Fault Circuit Interruption (AFCI), which helps reduce the risk of DC electrical fires.

Since the PV modules continue to generate electricity whenever there is sunlight, the installation and commissioning work should only be carried out by qualified personnel. Proper safety practices such as wearing fall protection equipment, using lockout/tagout procedures, insulated tools and checking the wiring polarity must be followed. These practices comply with the safety requirements set by TNB and DOSH under the Occupational Safety and Health Act (OSHA) 1994. During normal operation, the system produces no noise or harmful emissions, and by reducing the use of fossil-fuel electricity, it also contributes to cleaner air for the surrounding community.

7.5 Legal and Regulatory Framework

The proposed GCPV system follows the legal and regulatory requirements in Malaysia. The system is designed to be connected to the electricity grid under the Net Energy Metering (NEM) programme managed by SEDA Malaysia. TNB is responsible for providing and commissioning the three-phase bidirectional meter, while the PV system is connected through the existing household distribution board.

According to the published NEM Rakyat guidelines, the maximum allowable inverter capacity for residential systems is 5 kWac for single-phase houses and 12.5 kWac for three-phase houses. Since the proposed system uses a 3 kWac three-phase inverter, it is well within the permitted limit. Kilowatt alternating current is referred to as kWac. Measuring the maximum power output that an inverter can supply to your home or power grid is most frequently employed in the solar and renewable energy industries.

The installation must also comply with the Electricity Supply Act 1990 and the Electricity Regulations 1994 under the supervision of the Energy Commission (Suruhanjaya Tenaga). Therefore, the system should be designed, installed and tested by competent or registered personnel. In addition, the selected PV modules and inverter comply with the relevant Malaysian Standards, including MS IEC 61215 and MS IEC 61730 for PV modules, MS IEC 62109 for inverter safety, and MS IEC 60364 together with MS 1837 for low-voltage and PV installations. Since government policies and NEM quotas may change over time, the homeowner and the Registered PV Service Provider should always confirm the latest programme requirements before submitting the application.

7.6 Cultural Impact

Besides providing technical and financial benefits, installing rooftop solar panels can also encourage a more sustainable lifestyle within the community. As more houses in Seremban and Negeri Sembilan adopt solar energy, people become more familiar with renewable energy and are more likely to support environmentally friendly practices. This helps build a community that is more aware of energy conservation and supports Malaysia's transition towards a low-carbon future.

The PV modules are installed flush with the existing roof, so the overall appearance of the house and neighbourhood is maintained. This helps improve public acceptance of rooftop solar systems. During the installation process, contractors should minimise noise and avoid disturbing neighbouring houses while respecting the privacy of the residents. Safety labels in both Malay and English should also be provided. Finally, homeowners should be given simple guidance on how to monitor the system, isolate it during emergencies and carry out basic maintenance so that the PV system can be operated safely throughout its service life.

8. CONCLUSION

A grid-connected photovoltaic system has been designed for a residential house in Taman Warisan Puteri, Seremban, with the objective of maximising the energy harvested from the available roof area. Applying the standard GCPV sizing methodology, the measured 4.17 m × 3.85 m roof plane was found to accommodate six Trina Solar Vertex S+ TSM-440NEG9R.28 modules, configured as a single series string of six modules (N_p = 1, N_s = 6) connected to one MPP tracker of a 3 kW Huawei SUN2000-3KTL-M1 three-phase inverter, matched to the premise's three-phase supply. The configuration was verified to remain within the inverter's voltage window and input current limit across the full 20–75 °C cell-temperature range.

The proposed GCPV system uses six Trina Solar 440 W PV modules with a Huawei SUN2000-3KTL-M1 three-phase inverter, giving a total installed capacity of 2.64 kWp. Based on the design calculations, the system is expected to generate approximately 3,333 kWh of electricity per year, with a specific yield of about 1,263 kWh/kWp and a performance ratio (PR) of approximately 0.82. The system is also estimated to reduce around 1.95 tonnes of CO_2 emissions every year, contributing to a cleaner environment.

A complete electrical design was developed, including DC and AC isolators, surge protection devices (SPD), and a proper earthing system to ensure safe and reliable operation. Besides meeting the technical requirements, the proposed design also promotes sustainability by recommending three sustainable materials, systems and processes, supporting five United Nations Sustainable Development Goals (UN SDGs), and providing positive benefits to society, the environment, health and safety, legal compliance, and the local community.

Before the system is installed, the proposed design should be verified through an on-site assessment by a qualified solar PV service provider. This assessment should confirm the roof structure, shading conditions, installation setbacks and compliance with the latest regulatory requirements. Overall, the proposed design successfully meets the objective of maximising the use of the available roof space while providing a safe, reliable, cost-effective and environmentally friendly rooftop solar PV system.

REFERENCES

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[4] Parliament of Malaysia, Electricity Supply Act 1990 (Act 447).

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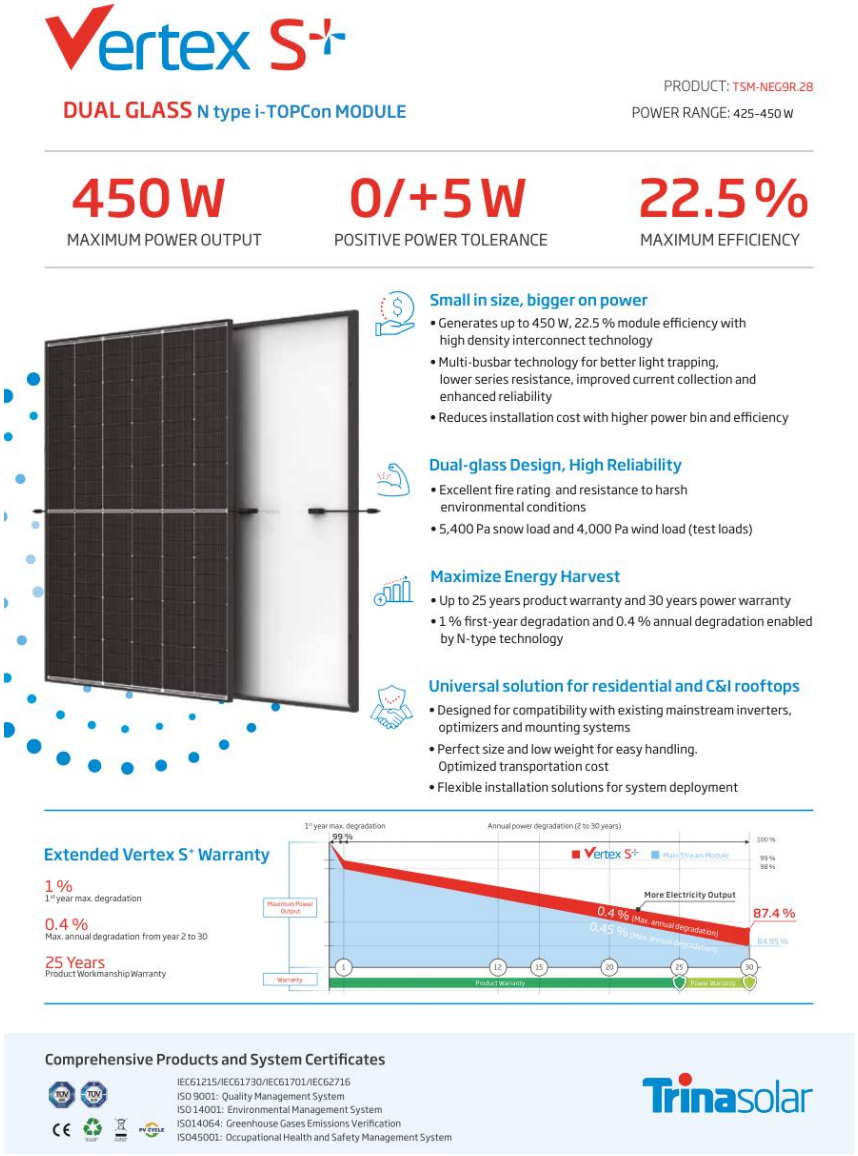
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APPENDICES

The manufacturer datasheets of the selected PV module and inverter are provided below, together with the site photographs used for the roof measurement and metering TNB box

Appendix A — PV Module Datasheet (Trina Solar Vertex S+ TSM-440NEG9R.28)

Trina Solar Vertex S+, TSM-440NEG9R.28, 440 Wp monocrystalline N-type i-TOPCon dual-glass module.

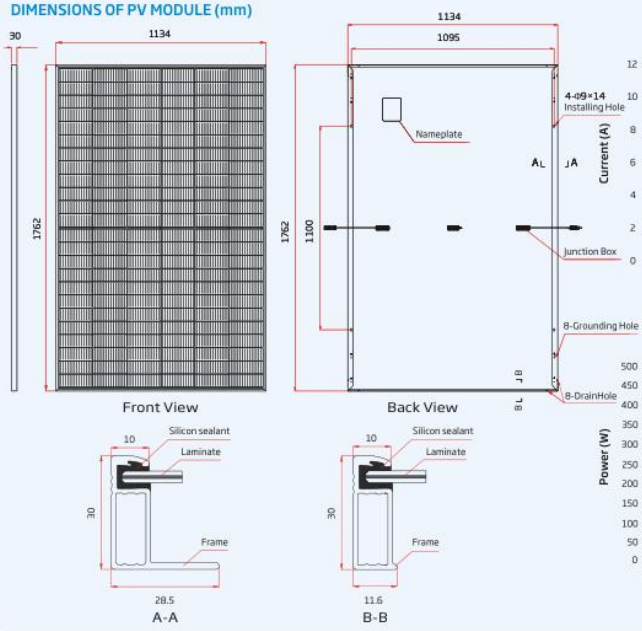


Appendix A.1: Trina Solar Vertex S+ datasheet, page 1.

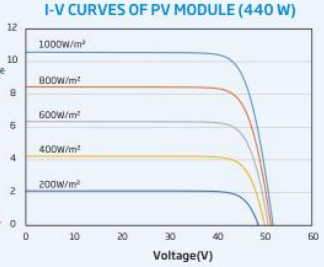
Vertex S+

DUAL GLASS N type i-TOPCon MODULE

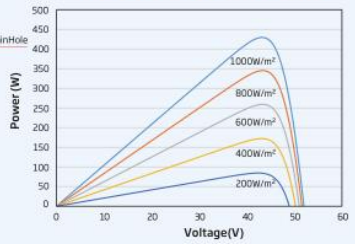
DIMENSIONS OF PV MODULE (mm)



I-V CURVES OF PV MODULE (440 W)



P-V CURVES OF PV MODULE (440 W)



ELECTRICAL DATA (STC)	TSM-425 NEG9R.28	TSM-430 NEG9R.28	TSM-435 NEG9R.28	TSM-440 NEG9R.28	TSM-445 NEG9R.28	TSM-450 NEG9R.28
Peak Power Watts-P_{MAX} (Wp)*	425	430	435	440	445	450
Power Tolerance-P_{MAX} (W)	0/+5					
Maximum Power Voltage-V_{MPP} (V)	42.9	43.2	43.6	44.0	44.3	44.6
Maximum Power Current-I_{MPP} (A)	9.92	9.96	9.99	10.01	10.05	10.09
Open Circuit Voltage-V_{OC} (V)	50.9	51.4	51.8	52.2	52.6	52.9
Short Circuit Current-I_{SC} (A)	10.56	10.59	10.64	10.67	10.71	10.74
Module Efficiency η m (%)	21.3	21.5	21.8	22.0	22.3	22.5

STC: Irradiance 1000 W/m², Cell Temperature 25 °C, Air Mass AM 1.5. *Measuring tolerance: ±3 %.

ELECTRICAL DATA (NOCT)	TSM-425 NEG9R.28	TSM-430 NEG9R.28	TSM-435 NEG9R.28	TSM-440 NEG9R.28	TSM-445 NEG9R.28	TSM-450 NEG9R.28
Maximum Power-P_{MAX} (Wp)	324	328	332	335	339	343
Maximum Power Voltage-V_{MPP} (V)	40.0	40.4	40.7	41.0	41.3	41.6
Maximum Power Current-I_{MPP} (A)	8.09	8.11	8.15	8.17	8.20	8.24
Open Circuit Voltage-V_{OC} (V)	48.2	48.7	49.1	49.4	49.8	50.1
Short Circuit Current-I_{SC} (A)	8.51	8.53	8.57	8.60	8.63	8.65

NOCT: Irradiance at 800 W/m², Ambient Temperature 20 °C, Wind Speed 1 m/s.

MECHANICAL DATA

Solar Cells	Monocrystalline
No. of cells	144 cells
Module Dimensions	1762×1134×30 mm
Weight	21.0 kg
Front Glass	1.6 mm, High Transmission, AR Coated Heat Strengthened Glass
Encapsulant material	POE/EVA
Back Glass	1.6 mm, Heat Strengthened Glass
Frame	30 mm Anodized Aluminium Alloy, Black
J-Box	IP 68 rated
Cables	Photovoltaic Technology Cable 4.0 mm² Landscape: 1100/1100 mm Portrait: 280/350 mm*
Connector	TS4 / MC4 EVO2*

*Special order only

TEMPERATURE RATINGS

NOCT (Nominal Operating Cell Temperature)	43 °C (±2 K)
Temperature Coefficient of P_{MAX}	-0.30 %/ K
Temperature Coefficient of V_{OC}	-0.24 %/ K
Temperature Coefficient of I_{SC}	0.04 %/ K

MAXIMUM RATINGS

Operational Temperature	-40 to +85 °C
Maximum System Voltage	1500 V DC (IEC)
Max Series Fuse Rating	20 A

WARRANTY

25 year Product Workmanship Warranty

30 year Power Warranty

1 % first year degradation

0.4 % Annual Power Attenuation

(Please refer to product warranty for details)

PACKAGING CONFIGURATION

Modules per box:	36 pieces
Modules per 40' container:	936 pieces

Trinasolar

CAUTION: READ SAFETY AND INSTALLATION INSTRUCTIONS BEFORE USING THE PRODUCT.

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Appendix A.2: Trina Solar Vertex S+ datasheet, page 2.

Appendix B — Inverter Datasheet (Huawei SUN2000-3KTL-M1)

Huawei SUN2000-3KTL-M1 — 3 kW three-phase residential string inverter (Smart Energy Controller).



Appendix B.1: Huawei SUN2000-M1 datasheet, page 1.

SUN2000-3/4/5/6/8/10KTL-M1 (High Current Version)

Technical Specification

Technical Specification	SUN2000 -3KTL-M1	SUN2000 -4KTL-M1	SUN2000 -5KTL-M1	SUN2000 -6KTL-M1	SUN2000 -8KTL-M1	SUN2000 -10KTL-M1
Efficiency						
Max. efficiency	98.2%	98.3%	98.4%	98.6%	98.6%	98.6%
European weighted efficiency	96.7%	97.1%	97.5%	97.7%	98.0%	98.1%
Input (PV)						
Recommended max. PV power [1]	4,500 Wp	6,000 Wp	7,500 Wp	9,000 Wp	12,000 Wp	15,000 Wp
Max. input voltage [2]	1,100 V					
Operating voltage range [3]	140–980 V					
Startup voltage	200 V					
Rated input voltage	600 V					
Max. input current per MPPT	13.5 A					
Max. short-circuit current	19.5 A					
Number of MPP trackers	2					
Max. input number per MPP tracker	1					
Input (DC Battery)						
Compatible battery	LUNA2000-5/10/15-S0, LUNA2000-7/14/21-S1					
Operating voltage range	600–980 V					
Max. operating current	16.7 A					
Max. charge power	10,000 W					
Max. discharge power	3300 W	4400 W	5500 W	6600 W	8800 W	10000 W
Output (On Grid)						
Grid connection	Three-phase					
Rated output power	3000 W	4000 W	5000 W	6000 W	8000 W	10,000 W
Max. apparent power	3300 VA	4400 VA	5500 VA	6600 VA	8800 VA	11,000 VA[4]
Rated output voltage	220 V AC/380 V AC, 230 V AC/400 V AC, 3W/N+PE					
Rated AC grid frequency	50 Hz/60 Hz					
Max. output current	5.1 A	6.8 A	8.5 A	10.1 A	13.5 A	16.9 A
Adjustable power factor	0.8 leading ... 0.8 lagging					
Max. total harmonic distortion	≤ 3%					
Output (Off Grid)						
BackupBox	BackupBox-B1					
Max. apparent power	3000 VA	3300 VA	3300 VA	3300 VA	3300 VA	3300 VA
Rated output voltage	220 V/230 V					
Max. output current	13.6 A	15 A	15 A	15 A	15 A	15 A
Power factor range	0.8 leading ... 0.8 lagging					
Protection Feature						
Input-side disconnection device	Yes					
Anti-islanding protection	Yes					
DC reverse polarity protection	Yes					
Insulation monitoring	Yes					
DC surge protection	Yes, compatible with TYPE II protection class according to EN/IEC 61643-11					
AC surge protection	Yes, compatible with TYPE II protection class according to EN/IEC 61643-11					
Residual current monitoring	Yes					
AC overcurrent protection	Yes					
AC short-circuit protection	Yes					
AC overvoltage protection	Yes					
Arc fault protection	Yes					
Ripple receiver control	Yes					
Battery charging from grid	Yes					
General Specification						
Operating temperature range	–25°C to +60°C (–13°F to +140°F)					
Relative operating humidity	0%–100% RH					
Max. operating altitude	4,000 m (13,123 ft.) (Derated above 2000 m)					
Cooling	Natural convection					
Display	LED Indicators; Integrated WLAN + FusionSolar app					
Communication	RS485; WLAN/Ethernet via Smart Dongle-WLAN-FE; 4G/3G/2G via Smart Dongle-4G (Optional)					
Weight (incl. mounting brackets)	17 kg (37.5 lb)					
Dimensions (incl. mounting brackets)	525 mm x 470 mm x 146.5 mm (20.7 in. x 18.5 in. x 5.8 in.)					
IP rating	IP65					
Nighttime power	< 5.5 W					
Optimizer Compatibility						
DC MBUS compatible optimizer	SUN2000-450W-P2, SUN2000-600W-P					
Standards Compliance (More Available Upon Request)						
Safety	EN/IEC 62109-1, EN/IEC 62109-2, IEC 62116					
Grid connection standards	G98, G99, EN 50438, CEI 0-21, VDE-AR-N-4105, AS 4777, C10/11, ABNT, UTE C15-712, RD 1699, TOR D4, NRS 097-2-1, IEC61727, IEC62116, DEWA					

*1 The inverter max input PV power is 20,000 Wp when long strings are designed and connected to SUN2000-450W-P2 or SUN2000-600W-P power optimizers.
*2 The max. input voltage is the upper limit of the DC voltage. Any higher input DC voltage would probably damage the inverter.
*3 Any DC input voltage beyond the operating voltage range may result in inverter malfunction.
*4 C10/11: 10,000 VA

Disclaimer: The preceding values are measured by an internal laboratory of Huawei in a specific environment. The actual values may vary with products, software versions, usage conditions, and environmental factors.

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Appendix B.2 : Huawei SUN2000-M1 datasheet, page 2.

Appendix C — Site Photographs

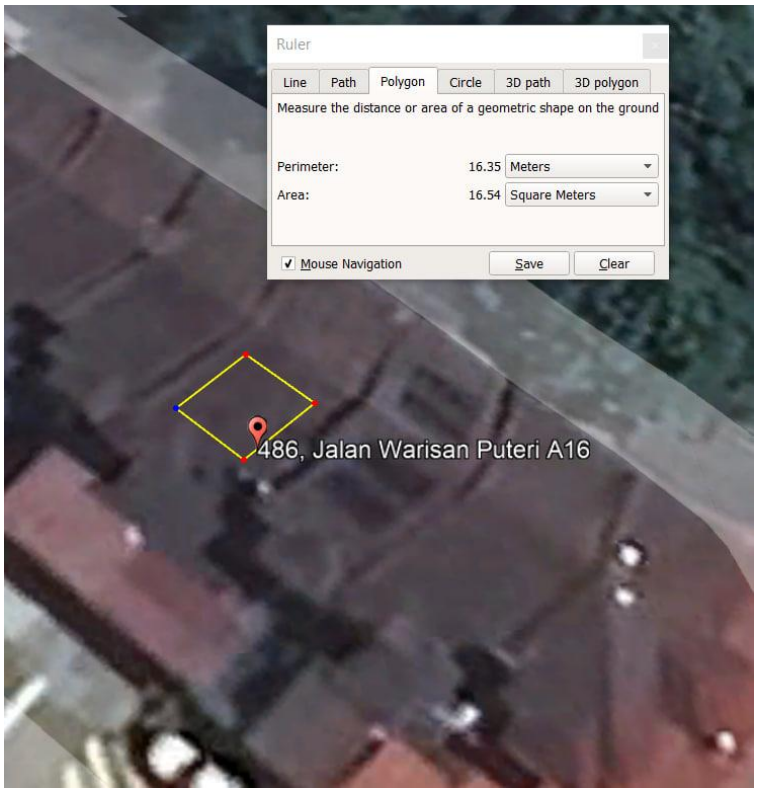


Figure C.1 : Roof polygon measurement: perimeter 16.35 m and area 16.54 m².

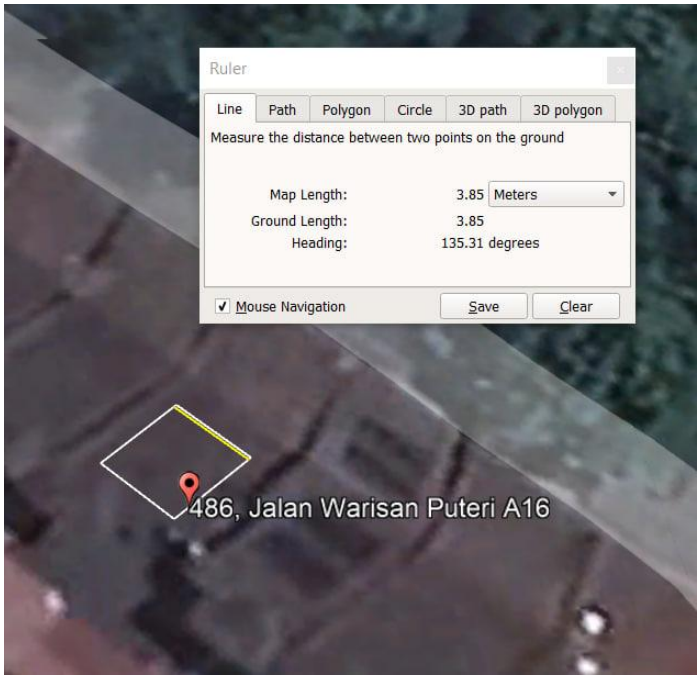


Figure C.2: Roof line measurement: 3.85 m.

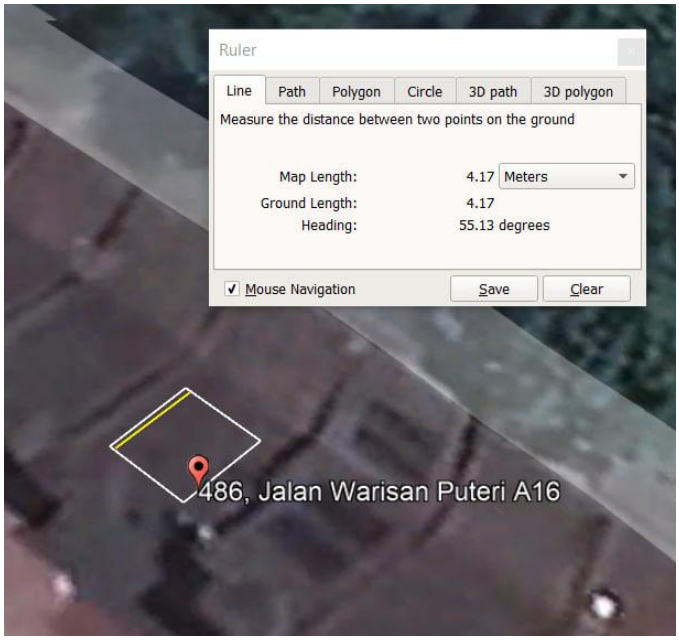


Figure C.3: Roof line measurement: 4.17 m.

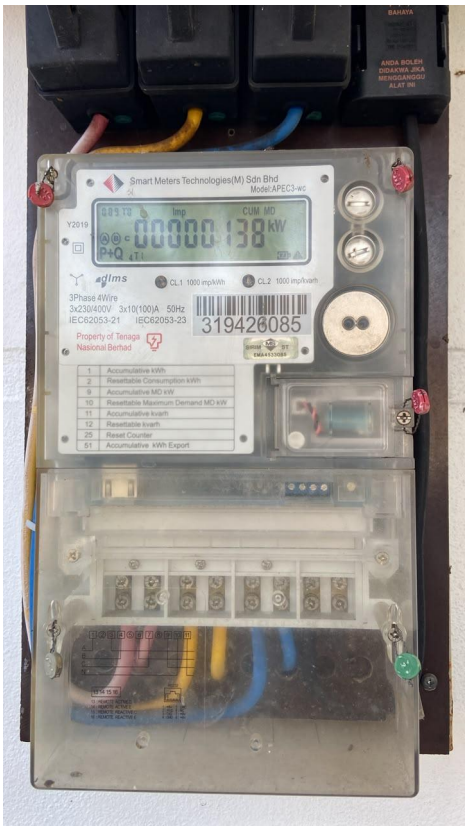


Figure C.4 : Existing TNB three-phase smart meter confirming the three-phase supply.